KERNEL-DEFINED MULTILINEAR OPERATORS AND CONDITIONAL ANALYTIC EXTENSIONS OF A FINITE PROJECTION THEORY

ELIUTH CHAVERO JASSO

Abstract. A companion article [3] studies finite composition trees of multilinear maps between finite-dimensional Hilbert spaces whose intermediate outputs are orthogonally projected onto prescribed subspaces, and proves an exact local error decomposition together with uniform coefficients k and $k-1$ for the ambient and projected root errors. The present article asks which parts of that setting survive when the finite-dimensional spaces are replaced by L^2 spaces and the multilinear maps by integral operators, and it is explicit at every step about what is proved, what is proved only under stated hypotheses, and what is merely proposed.

We prove that a kernel in $L^2(X^{a+1})$ defines a bounded multilinear operator on $L^2(X)$, and — this closes a gap in an earlier formulation — that the composite kernels obtained by inserting one such operator into another again lie in L^2, with $\|\kappa_L\|_{L^2(X^6)} \leq \|\kappa\|^2_{L^2(X^4)}$ in the ternary case, so that the associator of a kernel-defined operator is itself given by a square-integrable defect kernel. We prove that the associator characterises this defect: under separability of $L^2(X,\nu)$, the associator vanishes on all products of L^2 functions if and only if the defect kernel vanishes almost everywhere. We record the exact identity relating an associator to the commutator defect of left multiplication operators, and we are careful to present the resulting algebraic tensor as a definition rather than as a curvature. For finite cochain complexes we give the standard conditions under which a degree-zero graded operator descends to cohomology and commutes with a Hodge Laplacian, and we state the spectral-truncation version as a statement about the truncation only. An induced Markov operator and its Dirichlet form are presented as a conditional construction whose hypothesis we do not verify for any kernel. Continuum limits, membership in a pseudodifferential class, and microlocal regularity are open questions and are collected as such.

[2020]47H60, 47G10, 46E30, 58J50, 17A30 multilinear integral operator, associator kernel, orthogonal projection, cochain complex, Hodge Laplacian, spectral truncation, Markov operator

Contents

Date: 31 July 2026.

Companion to [3], which contains the finite-dimensional theory. Nothing in the present article has been checked by anyone other than the author, and no claim of originality is made.

1. Introduction

1.1. **What the companion article establishes.** Let $\mathcal{T}$ be a finite set of types, and for each $\tau \in \mathcal{T}$ let V_τ be a finite-dimensional Hilbert space, $Q_\tau : W_\tau \to V_\tau$ an isometry and $P_\tau = Q_\tau Q_\tau^*$ the associated orthogonal projector. A finite rooted ordered tree whose internal vertices carry multilinear maps $\mu_v : \prod_j V_{\tau(v,j)} \to V_{\tau(v)}$ admits two evaluations: the unprojected one, and the one in which the output of every internal vertex is replaced by its orthogonal projection. Writing E_T^{amb}, E_T^{proj}, $E_T^{\perp}$ and E_T^{red} for the four resulting measures of discrepancy at the root, the companion article proves that

$$\left(E_T^{\mathrm{amb}}\right)^2 = \left(E_T^{\mathrm{proj}}\right)^2 + \left(E_T^{\perp}\right)^2, \qquad E_T^{\mathrm{red}} = E_T^{\mathrm{proj}},$$

that the discrepancy at each vertex admits an exact decomposition into a locally created residual and a term for every nonempty subset of the children carrying an error, and that under uniform bounds $\|\mu_v\|_{\mathrm{op}} \le M$ and $\|r_v\|_{\mathrm{op}} \le \rho$ a tree with $k \ge 1$ internal vertices satisfies

$$E_T^{\mathrm{amb}} \le k\rho M^{k-1} L_T, \qquad E_T^{\mathrm{proj}} = E_T^{\mathrm{red}} \le (k-1)\rho M^{k-1} L_T, \tag{1}$$

L_T being the product of the leaf norms. The coefficient $k-1$ is an upper bound; the companion article does not show it is attained at fixed $\rho/M > 0$, and leaves the exact constant open already for two internal vertices.

We reproduce none of these proofs. They are stated above only to fix the notation and the hypotheses that the present article refers to, and every use of them below is a citation of [3].

1.2. **What this article does.** The finite theory has three features that are not obviously stable under passage to infinite dimensions: it uses operator norms of multilinear maps, it uses orthogonal projectors with closed range, and it uses composition of multilinear maps. Each raises a separate analytic question, and the answers are of different quality.

Section 2 shows that multilinear operators defined by square-integrable kernels behave well: they are bounded, they depend continuously on the kernel, and — Lemma 2.4 — their composites are again defined by square-integrable kernels. This last point matters because the earlier formulation of this material defined the associator kernel without establishing that it exists, and the estimate supplied here removes that gap.

Section 3 treats the associator. Proposition 3.4 shows that the associator determines the defect kernel exactly, under a separability hypothesis that we state explicitly because it is not automatic. Section 4 records the identity relating the associator to the commutator defect of left multiplication operators. We emphasise that the algebraic tensor obtained from an associator is a definition, not a curvature tensor of a connection.

Sections 5 and 6 concern structures that are finite or truncated: a variational formulation on a Stiefel manifold, and the descent of a degree-zero graded operator to the cohomology of a finite cochain complex, together with its spectral-truncation analogue. These are standard statements; we give them because the truncated versions are what the computational work of the companion program actually tests, and it is important that they not be mistaken for continuum statements.

Section 7 presents an induced Markov operator and its Dirichlet form as a *conditional construction.* Its hypothesis — that a contraction of the kernel produces a symmetric nonnegative weight — is not verified here for any kernel, and we say so.

Section 9 collects the open analytical questions. Continuum limits of the finite theory, membership of kernel-defined operators in a multilinear pseudodifferential class, and microlocal regularity are questions, not results, and no claim about them is made anywhere in this article.

2. Multilinear operators defined by square-integrable kernels

Let (X,ν) be a σ-finite measure space and $H = L^2(X,\nu)$. Throughout, ν is the measure; the letter μ is reserved for the multilinear maps of the finite theory and does not appear in this section.

Definition 2.1 (Kernel-defined multilinear operator)**.** For $\kappa \in L^2(X^{a+1}, \nu^{\otimes(a+1)})$ define

$$\mathcal{K}_\kappa(f_1,\dots,f_a)(p) = \int_{X^a} \kappa(p;q_1,\dots,q_a) \prod_{j=1}^{a} f_j(q_j)\, d\nu^{\otimes a}(q_1,\dots,q_a).$$

Proposition 2.2 (Boundedness)**.** *If $\kappa \in L^2(X^{a+1})$ then $\mathcal{K}_\kappa$ is a well-defined bounded a-linear map $H^a \to H$, and*

$$\|\mathcal{K}_\kappa(f_1,\dots,f_a)\|_2 \le \|\kappa\|_{L^2(X^{a+1})} \prod_{j=1}^{a} \|f_j\|_2.$$

Proof. By Tonelli's theorem, $p \mapsto \|\kappa(p;\cdot)\|_{L^2(X^a)}$ is measurable and

$$\int_X \|\kappa(p;\cdot)\|^2_{L^2(X^a)}\, d\nu(p) = \|\kappa\|^2_{L^2(X^{a+1})}.$$

For ν-almost every p the function $\kappa(p;\cdot)$ lies in $L^2(X^a)$, and $f_1 \otimes \cdots \otimes f_a \in L^2(X^a)$ with norm $\prod_j \|f_j\|_2$ by Tonelli again. Cauchy–Schwarz on X^a gives, for almost every p, both absolute convergence of the defining integral and

$$|\mathcal{K}_\kappa(f_1,\dots,f_a)(p)| \le \|\kappa(p;\cdot)\|_{L^2(X^a)} \prod_j \|f_j\|_2.$$

Squaring and integrating in p gives the stated estimate. ☐

Proposition 2.3 (Dependence on the kernel)**.** *For $\kappa, \widetilde{\kappa} \in L^2(X^{a+1})$,*

$$\|\mathcal{K}_\kappa(f_1,\dots,f_a) - \mathcal{K}_{\widetilde{\kappa}}(f_1,\dots,f_a)\|_2 \le \|\kappa - \widetilde{\kappa}\|_{L^2(X^{a+1})} \prod_j \|f_j\|_2.$$

In particular $\kappa_n \to \kappa$ in $L^2(X^{a+1})$ implies $\mathcal{K}_{\kappa_n} \to \mathcal{K}_\kappa$ in the multilinear operator norm.

Proof. $\kappa \mapsto \mathcal{K}_\kappa$ is linear, so this is Proposition 2.2 applied to $\kappa - \widetilde{\kappa}$. The second assertion follows because the estimate is uniform over unit vectors. ☐

2.1. Composite kernels.

We now compose two ternary operators; the general a-ary case is identical.

Lemma 2.4 (Composite kernels are square-integrable)**.** *Let $\kappa \in L^2(X^4)$ and define, for almost every argument,*

$$\kappa_L(p;q_1,\dots,q_5) = \int_X \kappa(p;u,q_4,q_5)\,\kappa(u;q_1,q_2,q_3)\, d\nu(u),$$
$$\kappa_R(p;q_1,\dots,q_5) = \int_X \kappa(p;q_1,q_2,u)\,\kappa(u;q_3,q_4,q_5)\, d\nu(u).$$

Then the defining integrals converge absolutely for $\nu^{\otimes 6}$-almost every $(p;q_1,\dots,q_5)$, both κ_L and κ_R belong to $L^2(X^6)$, and

$$\|\kappa_L\|_{L^2(X^6)} \le \|\kappa\|^2_{L^2(X^4)}, \qquad \|\kappa_R\|_{L^2(X^6)} \le \|\kappa\|^2_{L^2(X^4)}.$$

Proof. Write $g(p, q_4, q_5) = \|\kappa(p; \cdot, q_4, q_5)\|_{L^2(X)}$ and $h(q_1, q_2, q_3) = \|\kappa(\cdot; q_1, q_2, q_3)\|_{L^2(X)}$, both measurable by Tonelli. Cauchy–Schwarz in the variable u gives, pointwise,

$$\int_X |\kappa(p; u, q_4, q_5)\kappa(u; q_1, q_2, q_3)|\, d\nu(u) \le g(p, q_4, q_5)\, h(q_1, q_2, q_3),$$

so the integral defining κ_L converges absolutely whenever the right-hand side is finite, and

$$|\kappa_L(p; q_1, \dots, q_5)|^2 \le g(p, q_4, q_5)^2\, h(q_1, q_2, q_3)^2.$$

Integrating over X^6 and using Tonelli to separate the variables,

$$\|\kappa_L\|^2_{L^2(X^6)} \le \Big(\int_{X^3} g^2\, d\nu^{\otimes 3}\Big)\Big(\int_{X^3} h^2\, d\nu^{\otimes 3}\Big) = \|\kappa\|^2_{L^2(X^4)} \cdot \|\kappa\|^2_{L^2(X^4)},$$

since each factor is $\|\kappa\|^2_{L^2(X^4)}$ by Tonelli. Finiteness of the left-hand side forces the right-hand side of the pointwise estimate to be finite almost everywhere, which is the asserted almost-everywhere absolute convergence. The argument for κ_R is the same with the roles of the variables exchanged. $\square$

Remark 2.5. Lemma 2.4 is what makes Definition 3.2 below legitimate. An earlier formulation of this material introduced κ_L, κ_R and their difference, and defined an energy as the squared $L^2(X^6)$ norm of that difference, without establishing that any of the three exists. Square-integrability of a kernel does not in general pass to composites of the associated operators for free; here it does, and the reason is the elementary estimate above.

3. The associator of a kernel-defined operator

For a ternary operation there is no canonical associator without a choice of how the compositions are inserted. We fix one, and use it consistently.

Definition 3.1 (Five-input ternary associator)**.** For a ternary multilinear map μ on a vector space,

$$A^{(5)}_\mu(x_1, \dots, x_5) = \mu(\mu(x_1, x_2, x_3), x_4, x_5) - \mu(x_1, x_2, \mu(x_3, x_4, x_5)).$$

In operadic notation, $A^{(5)}_\mu = \mu \circ_1 \mu - \mu \circ_3 \mu$ [13, 10].

Other conventions — the binary associator induced by anchoring one argument, or insertion at another slot — have different domains and different meanings. They must not be merged into a single quantity, and we do not do so.

Definition 3.2 (Defect kernel and associator energy)**.** For $\kappa \in L^2(X^4)$ put $\Phi_\kappa = \kappa_L - \kappa_R$, which lies in $L^2(X^6)$ by Lemma 2.4, with $\|\Phi_\kappa\|_{L^2(X^6)} \le 2\|\kappa\|^2_{L^2(X^4)}$. The *associator energy* is

$$\mathfrak{E}_A(\kappa) = \|\Phi_\kappa\|^2_{L^2(X^6)} \ \le\ 4\|\kappa\|^4_{L^2(X^4)}.$$

Proposition 3.3 (The defect kernel represents the associator)**.** *For all $f_1, \dots, f_5 \in L^2(X)$,*

$$A^{(5)}_{\mathcal{K}_\kappa}(f_1, \dots, f_5)(p) = \int_{X^5} \Phi_\kappa(p; q_1, \dots, q_5) \prod_{j=1}^{5} f_j(q_j)\, d\nu^{\otimes 5}(q)$$

for ν-almost every p; that is, $A^{(5)}_{\mathcal{K}_\kappa} = \mathcal{K}_{\Phi_\kappa}$.

Proof. By Lemma 2.4 the function $(u, q) \mapsto \kappa(p; u, q_4, q_5)\kappa(u; q_1, q_2, q_3) \prod_j f_j(q_j)$ is absolutely integrable over X^6 for almost every p: indeed Cauchy–Schwarz bounds its integral by $\|\kappa(p; \cdot)\|_{L^2(X^3)} \|\kappa\|_{L^2(X^4)} \prod_j \|f_j\|_2$, which is finite for almost every p. Fubini's theorem therefore permits the interchange of the u-integration with the q-integration, which converts $\mathcal{K}_\kappa(\mathcal{K}_\kappa(f_1, f_2, f_3), f_4, f_5)$ into integration against κ_L. The same argument converts $\mathcal{K}_\kappa(f_1, f_2, \mathcal{K}_\kappa(f_3, f_4, f_5))$ into integration against κ_R. Subtracting gives the claim. $\square$

Proposition 3.4 (The associator determines the defect kernel)**.** *Assume in addition that $L^2(X,\nu)$ is separable. Then*

$$A^{(5)}_{\mathcal{K}_\kappa}(f_1,\dots,f_5)=0 \textit{ in } L^2(X) \textit{ for all } f_1,\dots,f_5\in L^2(X) \iff \Phi_\kappa=0 \quad \nu^{\otimes 6}\textit{-a.e.}$$

Proof. If $\Phi_\kappa=0$ almost everywhere then the associator vanishes by Proposition 3.3.

Conversely, let $\{g_m\}_{m\in\mathbb{N}}$ be a countable dense subset of $L^2(X,\nu)$, which exists by separability. For each of the countably many 5-tuples $(m_1,\dots,m_5)$, Proposition 3.3 and the hypothesis give

$$\int_{X^5}\Phi_\kappa(p;q)\prod_{j=1}^{5}g_{m_j}(q_j)\,d\nu^{\otimes 5}(q)=0$$

for p outside a ν-null set $N_{m_1\cdots m_5}$. Let $N=\bigcup N_{m_1\cdots m_5}$, a countable union of null sets, hence null. Enlarging N if necessary, we may also assume that $\Phi_\kappa(p;\cdot)\in L^2(X^5)$ for $p\notin N$, which holds almost everywhere by Lemma 2.4 and Tonelli.

Fix $p\notin N$. The linear span of $\{g_{m_1}\otimes\cdots\otimes g_{m_5}\}$ is dense in $L^2(X^5)$: finite linear combinations of products of elements of a dense subset of $L^2(X)$ are dense in the Hilbert space tensor product $L^2(X)^{\otimes 5}\cong L^2(X^5)$ [11, Sec. II.4]. The function $\Phi_\kappa(p;\cdot)$ is an element of $L^2(X^5)$ orthogonal to a dense subspace, hence zero. As this holds for every $p\notin N$, we get $\Phi_\kappa=0$ almost everywhere by Tonelli. □

Remark 3.5. Separability is a genuine hypothesis and is not automatic for a σ-finite measure space; it holds when the underlying σ-algebra is countably generated modulo null sets. Without it the countable-dense-family argument is unavailable, and we make no claim.

4. Associators and commutator defects

Let $\circ$ be any bilinear operation on a vector space V — no algebra axioms are assumed. Write $\mathsf{L}_x z=x\circ z$ for left multiplication, $[x,y]=x\circ y-y\circ x$, and

$$A(x,y,z)=(x\circ y)\circ z-x\circ(y\circ z),\qquad R_{\mathrm{std}}(x,y)=[\mathsf{L}_x,\mathsf{L}_y]-\mathsf{L}_{[x,y]}.$$

Theorem 4.1 (Commutator defect and associator)**.** *For all $x,y,z\in V$,*

$$R_{\mathrm{std}}(x,y)z=A(y,x,z)-A(x,y,z).$$

Proof. Expanding the definitions,

$$\begin{aligned}R_{\mathrm{std}}(x,y)z&=x\circ(y\circ z)-y\circ(x\circ z)-(x\circ y)\circ z+(y\circ x)\circ z\\&=\big[x\circ(y\circ z)-(x\circ y)\circ z\big]-\big[y\circ(x\circ z)-(y\circ x)\circ z\big]\\&=-A(x,y,z)+A(y,x,z).\end{aligned}$$

□

Definition 4.2 (Associator-based algebraic tensor)**.** One may *define*, by convention, $R_{\mathrm{alg}}:=A$.

Remark 4.3 (This is a definition, not a curvature theorem). Definition 4.2 assigns a name to the associator. It does not identify R_{alg} with the curvature tensor of a connection: no connection, no bundle and no covariant derivative have been introduced, and none is implied. The only universal identity available is the antisymmetrised relation of Theorem 4.1. An outright identity $R_{\mathrm{std}}=A$ would require additional hypotheses, particular symmetries, or a different convention, and we do not assert one.

Remark 4.4 (Distinct identities). Associativity, cyclic symmetry, the Jacobi identity and Filippov's fundamental identity [6] are distinct conditions. A small residual for one of them implies nothing about the others, and residuals for different conventions must not be aggregated into a single score. The companion article's numerical study of signed combinations [3] keeps them separate for this reason.

5. Reduced operators, closure leakage, and a variational formulation

5.1. **Reduced operators.** Let W be a Hilbert space, $Q : W \to V$ an isometry and $P = QQ^*$. For a multilinear map μ on V the *reduced map* is $\bar{\mu}(z_1, \ldots, z_a) = Q^*\mu(Qz_1, \ldots, Qz_a)$, with ambient realisation $Q\bar{\mu}(z_1, \ldots, z_a) = P\mu(Qz_1, \ldots, Qz_a)$. The subspace family is invariant under μ exactly when the *closure-leakage map* $\mathcal{C}_\mu = (I - P)\mu(P\,\cdot, \ldots, P\,\cdot)$ vanishes identically; its operator norm $\rho_\mu = \|\mathcal{C}_\mu\|_{\mathrm{op}}$ is the quantity denoted ρ in (1).

Remark 5.1 (An operator norm is not a sampled average). Numerical work frequently reports a normalised sampled quantity such as

$$\mathfrak{E}_{\text{closure}} = \mathbb{E}\left[\frac{\|(I-P)\mu(PX_1, \ldots, PX_a)\|^2}{\|\mu(PX_1, \ldots, PX_a)\|^2 + \varepsilon}\right]$$

for random X_i. This is not ρ_μ, and a small value of it does not bound ρ_μ. The two must be reported separately.

5.2. **Low-rank representation.** To avoid storing a coefficient tensor of size $d_0 d_1 \cdots d_a$ one may use a rank-R canonical polyadic representation [8]. Such a representation is not unique: factors may be rescaled subject to a product constraint, and components permuted, without changing the tensor. Any comparison of factors must therefore be carried out modulo scaling, phase and permutation, and identifiability requires hypotheses of Kruskal type [9]. A low-rank representation is a computational device; on its own it introduces no algebraic structure.

5.3. **Variational formulation on a Stiefel manifold.** Parametrising $P = QQ^*$ with $Q^*Q = I_r$ places Q on the Stiefel manifold

$$\mathrm{St}(d,r) = \{Q \in \mathbb{K}^{d\times r} : Q^*Q = I_r\}, \qquad T_Q\mathrm{St}(d,r) = \{Z : Q^*Z + Z^*Q = 0\}.$$

With respect to the metric induced by the Euclidean embedding, the Riemannian gradient of a smooth objective $\mathfrak{E}$ with Euclidean gradient G is

$$\operatorname{grad}\mathfrak{E}(Q) = G - Q\,\mathrm{sym}(Q^*G), \qquad \mathrm{sym}(B) = \tfrac{1}{2}(B + B^*),$$

see [5, Sec. 2.4] and [1, Ch. 3]. The canonical metric on $\mathrm{St}(d,r)$ gives a different gradient; we state the embedded Euclidean one because that is the one used in the accompanying computations.

Remark 5.2. A numerically located stationary point of such an objective need not be a global minimum, and the subspace it determines need not be canonical in any sense. This is the ordinary situation for nonconvex optimisation on a manifold, and no result below depends on the contrary.

The tangent space $T_Q\mathrm{St}(d,r)$ is the one place in this article where the word *tangent* refers to a tangent space. Elsewhere, $\operatorname{ran} P$ is a fixed linear subspace and the words *projected* and *orthogonal* are used.

6. Finite cochain complexes and spectral truncation

6.1. **Descent to cohomology.** Let $C^\bullet : C^0 \xrightarrow{d_0} C^1 \xrightarrow{d_1} \cdots \to C^n$ be a finite complex of finite-dimensional inner-product spaces with $d_{p+1}d_p = 0$, and $H^p(C,d) = \ker d_p / \operatorname{im} d_{p-1}$.

Proposition 6.1 (Descent). *Let $S = (S_p)_p$ be a graded operator* of degree zero, *i.e.* $S_p : C^p \to C^p$ *for each p. If $S_{p+1}d_p = d_pS_p$ for every p, then S induces a well-defined map $S_* : H^p(C,d) \to H^p(C,d)$ for every p.*

Proof. If $x \in \ker d_p$ then $d_pS_px = S_{p+1}d_px = 0$, so S_p preserves cocycles. If $x = d_{p-1}y$ then $S_px = S_pd_{p-1}y = d_{p-1}S_{p-1}y$, so S_p preserves coboundaries. Hence $[x] \mapsto [S_px]$ is well defined. □

The degree-zero hypothesis is required and is stated explicitly; without it the two displayed compatibilities do not even typecheck.

6.2. **Hodge compatibility.** With d^* the adjoints and $\Delta_p = d_{p-1}d_{p-1}^* + d_p^* d_p$ the Hodge Laplacian, put $\mathcal{H}^p = \ker \Delta_p$.

Proposition 6.2 (Harmonic subspaces)**.** *If S commutes with d and with d^* in the graded sense, then S commutes with Δ, hence $S_p(\mathcal{H}^p) \subseteq \mathcal{H}^p$.*

Proof. Immediate from $\Delta = dd^* + d^*d$ and bilinearity of the commutator. The last assertion follows because $\Delta_p S_p x = S_p \Delta_p x = 0$ for $x \in \mathcal{H}^p$. □

For a finite complex, $\mathcal{H}^p \cong H^p(C, d)$ by the finite-dimensional Hodge decomposition; on a compact Riemannian manifold the corresponding statement for the de Rham complex is the Hodge theorem [12, Ch. 6]. The two settings are different and we do not pass from one to the other.

6.3. **Spectral truncation.** Let Δ now be the Hodge Laplacian of a compact Riemannian manifold, let $\Pi_{\le\Lambda}$ be its spectral projector onto eigenvalues at most Λ, and let $V_\Lambda = \operatorname{ran} \Pi_{\le\Lambda}$, a finite-dimensional space. For a family of operators $S^\kappa_{e_i,e_j}$ obtained from a kernel by fixing two of its arguments, define

$$\mathfrak{E}_d^\Lambda(\kappa) = \sum_{i,j} \Big(\|[d, S^\kappa_{e_i,e_j}]\Pi_{\le\Lambda}\|^2_{\mathrm{HS}} + \|[d^*, S^\kappa_{e_i,e_j}]\Pi_{\le\Lambda}\|^2_{\mathrm{HS}} \Big).$$

Proposition 6.3 (Truncated compatibility)**.** *If $\mathfrak{E}_d^\Lambda(\kappa) = 0$ then every $S^\kappa_{e_i,e_j}$ commutes with d and d^* on V_Λ, and therefore descends to the cohomology of the truncated complex $\Pi_{\le\Lambda}C^\bullet$ by Proposition 6.1.*

Remark 6.4 (The truncation is not the continuum)*.* Proposition 6.3 is a statement about V_Λ only. It gives no information about $[d, S^\kappa]$ on the orthogonal complement of V_Λ, and therefore none about the untruncated complex. Vanishing of $\mathfrak{E}_d^\Lambda$ for every Λ would be a different and much stronger hypothesis. Furthermore, a numerically located local minimum of $\mathfrak{E}_d^\Lambda$ does not imply $\mathfrak{E}_d^\Lambda = 0$; that conclusion requires an additional surjectivity or controllability hypothesis on the variations of the defect map, which we do not have.

7. An induced Markov operator: a conditional construction

This section is a construction under a hypothesis that is *not verified here for any kernel.* It is included because it is the shape a diffusion-theoretic extension would take, and because stating it precisely makes clear what would have to be proved.

Construction 7.1 (Induced Markov operator)**.** Suppose that a declared contraction of a kernel produces a measurable $\mathcal{W} : X \times X \to [0, \infty)$ with $\mathcal{W}(p, s) = \mathcal{W}(s, p)$, and set

$$\mathsf{d}(p) = \int_X \mathcal{W}(p, s)\, d\nu(s).$$

Assume $0 < \mathsf{d}(p) < \infty$ for ν-almost every p. Define $\mathcal{P}(p, s) = \mathcal{W}(p, s)/\mathsf{d}(p)$, the measure $d\lambda = \mathsf{d}\, d\nu$, the operator $(\mathcal{P}f)(p) = \int_X \mathcal{P}(p, s)f(s)\, d\nu(s)$, and $\Delta_{\mathcal{P}} = I - \mathcal{P}$.

Proposition 7.2 (Self-adjointness and positivity)**.** *Under the hypotheses of Construction 7.1, $\mathcal{P}$ is self-adjoint on $L^2(X, \lambda)$, and for $f \in L^2(X, \lambda)$*

$$\langle f, \Delta_{\mathcal{P}} f\rangle_{L^2(\lambda)} = \tfrac{1}{2} \iint_{X\times X} |f(p) - f(s)|^2\, \mathcal{W}(p, s)\, d\nu(p)\, d\nu(s) \ \ge\ 0.$$

Proof. For $f, g \in L^2(X, \lambda)$,

$$\langle \mathcal{P}f, g\rangle_{L^2(\lambda)} = \int_X \Big(\int_X \frac{\mathcal{W}(p, s)}{\mathsf{d}(p)} f(s)\, d\nu(s) \Big) \overline{g(p)}\, \mathsf{d}(p)\, d\nu(p) = \iint \mathcal{W}(p, s) f(s)\overline{g(p)}\, d\nu(s)\, d\nu(p),$$

which is symmetric in the pair (f,g) after exchanging $p \leftrightarrow s$ and using $\mathcal{W}(p,s) = \mathcal{W}(s,p)$; hence $\mathcal{P}$ is self-adjoint. Expanding,

$$\tfrac{1}{2}\iint |f(p)-f(s)|^2 \mathcal{W}(p,s) = \int_X |f(p)|^2 \mathsf{d}(p)\, d\nu(p) - \operatorname{Re}\iint \mathcal{W}(p,s) f(s)\overline{f(p)},$$

using symmetry of $\mathcal{W}$ for both cross terms, and the right-hand side is $\langle f,f\rangle_{L^2(\lambda)} - \langle \mathcal{P}f, f\rangle_{L^2(\lambda)}$, which is $\langle f, \Delta_{\mathcal{P}} f\rangle_{L^2(\lambda)}$. □

The operator $\Delta_{\mathcal{P}}$ is the graph Laplacian of the weight $\mathcal{W}$; the Dirichlet form above is the standard one, and the diffusion-map construction [4] is the analytic setting in which such operators are usually studied.

Remark 7.3 (Spectral dimension is a further hypothesis). If $e^{-t\Delta_{\mathcal{P}}}$ is trace class one may set $\Theta(t) = \operatorname{Tr} e^{-t\Delta_{\mathcal{P}}}$, and *if* $\Theta(t) \sim Ct^{-d_s/2}$ as $t \downarrow 0$ one may call $d_s = -2\lim_{t\downarrow 0} \frac{d\log\Theta(t)}{d\log t}$ the spectral dimension. Both the trace-class property and the existence of the limit are additional hypotheses. Neither is a consequence of a kernel being square-integrable.

Remark 7.4 (The hypothesis of Construction 7.1 is unverified). We exhibit no contraction of a kernel $\kappa \in L^2(X^4)$ that is guaranteed to produce a symmetric nonnegative $\mathcal{W}$ with $0 < \mathsf{d} < \infty$ almost everywhere, and we give no condition on κ under which one exists. Until such a condition is supplied, Construction 7.1 has no verified instance.

8. Multiscale transport defects

Let (V_N, μ_N, P_N) be a family indexed by a resolution parameter N, with prolongation maps $J_N^M : V_N \to V_M$ and restriction maps $R_M^N : V_M \to V_N$. Define the *transport defects*

$$\mathcal{D}_\mu^{N,M} = J_N^M \mu_N - \mu_M(J_N^M\cdot, \dots, J_N^M\cdot), \qquad \mathcal{D}_P^{N,M} = P_M J_N^M - J_N^M P_N,$$

and compare subspaces by $\|P_M - J_N^M P_N R_M^N\|$ or by principal angles.

These are definitions. No theorem is asserted about them here.

Remark 8.1 (A decreasing sequence is not a limit theorem). A finite decreasing sequence of transport defects does not establish a continuum limit. Establishing one would additionally require: explicit topologies on the family; uniform bounds on $\|\mu_N\|_{\mathrm{op}}$, on $\|P_N\|$ and on the leakage norms ρ_N; a compactness argument; convergence of the maps; convergence of the projectors in a stated operator topology; stability of whichever identities are of interest; and identification of the limiting operator. See Question 9.1.

9. Open analytical questions

The following are questions. None of them is a result of this article, and nothing above depends on any of them.

Question 9.1 (Continuum limit). Under what hypotheses on a family (V_N, μ_N, P_N) with prolongations J_N^M do the unprojected and recursively projected evaluations of a fixed finite tree converge, and does the estimate $E_T^{\mathrm{proj}} \le (k-1)\rho M^{k-1} L_T$ of (1) pass to the limit? A plausible set of hypotheses is: $J_N^M \mu_N(R\,\cdot, \dots, R\,\cdot) \to \mu$ and $J_N P_N R_N \to P$ in a stated topology, together with $\sup_N \|\mu_N\|_{\mathrm{op}} < \infty$, $\sup_N \|P_N\| \le 1$ and $\sup_N \rho_N < \infty$. For trees whose size grows with N a further control on depth and on the summability of the sources would be needed.

Question 9.2 (Membership in a multilinear pseudodifferential class). Under what hypotheses does a kernel-defined operator $\mathcal{K}_\kappa$ admit a local symbol $a(x, \xi_1, \dots, \xi_a)$ satisfying estimates of the form

$$|\partial_x^\alpha \partial_{\xi_1}^{\beta_1} \cdots \partial_{\xi_a}^{\beta_a} a| \le C_{\alpha,\beta}\Big(1 + \textstyle\sum_j |\xi_j|\Big)^{m - \varrho|\beta| + \delta|\alpha|},$$

so that it belongs to a multilinear Hörmander class? Square-integrability of κ is far from sufficient. The multilinear Calderón–Zygmund theory of Grafakos and Torres [7] and the bilinear Hörmander classes of Bényi, Maldonado, Naibo and Torres [2] indicate what such a theory requires: symbol classes, invariance under changes of coordinates, Sobolev continuity, composition, adjoints, and stability of the class under the formation of associators. A realistic first target would be a compact manifold, kernels smooth off the diagonal, symbols of order zero and spectral projectors as the P_τ.

Question 9.3 (Microlocal regularity). If Question 9.2 were answered affirmatively, what containment of the form $\mathrm{WF}(\mathcal{K}_\kappa(f_1,\dots,f_a)) \subseteq \mathcal{R}_\kappa(\mathrm{WF}(f_1),\dots,\mathrm{WF}(f_a))$ holds, for which canonical relation $\mathcal{R}_\kappa$? Does orthogonal projection create singularities; is the orthogonal component of the error more regular than the projected one; and is there a microlocal analogue of the passage from k to $k-1$ sources?

Question 9.4 (Effect of projection on the class). Is a class of kernel-defined operators stable under the operation $\mathcal{S} \mapsto P\mathcal{S}(P\,\cdot,\dots,P\,\cdot)$ that the finite theory performs at every vertex, and how does the leakage norm ρ behave under it?

Question 9.5 (Verifying Construction 7.1). Give a condition on $\kappa \in L^2(X^4)$ under which some declared contraction yields a symmetric nonnegative weight with $0 < \mathsf{d} < \infty$ almost everywhere.

We note explicitly that holonomic D-modules, the Riemann–Hilbert correspondence and algebraisation of a limiting projector have sometimes been mentioned in connection with this circle of ideas. No hypothesis, no statement and no proof relating them to the material above exists in this article, and they are therefore not listed even as questions.

10. Summary of status

Table 1. Status of each statement in this article.

Statement	Status
Prop. 2.2, 2.3	proved
Lemma 2.4	proved; closes a gap in an earlier formulation
Prop. 3.3	proved, using Lemma 2.4
Prop. 3.4	proved under separability of $L^2(X,\nu)$
Thm. 4.1	proved; an identity for any bilinear operation
Def. 4.2	a definition, not a theorem; see Remark 4.3
Prop. 6.1, 6.2	proved for finite complexes; standard
Prop. 6.3	proved for the truncation only; see Remark 6.4
Constr. 7.1, Prop. 7.2	conditional construction; hypothesis unverified
Remark 7.3	definition under two further hypotheses
Section 8	definitions only
Section 9	open questions
Eq. (1)	proved in [3]; not reproved here

None of the statements above has been checked by anyone other than the author, and no assessment of originality has been carried out for any of them. Several — notably Propositions 6.1 and 6.2, and the Stiefel gradient of Section 5 — are standard, and are included for completeness with citations rather than as contributions.

References

1. P.-A. Absil, R. Mahony, and R. Sepulchre, *Optimization algorithms on matrix manifolds*, Princeton University Press, 2008.

2. Árpád Bényi, Diego Maldonado, Virginia Naibo, and Rodolfo H. Torres, *On the Hörmander classes of bilinear pseudodifferential operators*, Integral Equations and Operator Theory **67** (2010), no. 3, 341–364.
3. Eliuth Chavero Jasso, *Error propagation under recursive orthogonal projection of finite multilinear composition trees*, 2026, Companion article, containing the finite-dimensional theory cited throughout.
4. Ronald R. Coifman and Stéphane Lafon, *Diffusion maps*, Applied and Computational Harmonic Analysis **21** (2006), no. 1, 5–30.
5. Alan Edelman, Tomás A. Arias, and Steven T. Smith, *The geometry of algorithms with orthogonality constraints*, SIAM Journal on Matrix Analysis and Applications **20** (1998), no. 2, 303–353.
6. V. T. Filippov, *n-Lie algebras*, Siberian Mathematical Journal **26** (1985), no. 6, 879–891, Translation of Sibirsk. Mat. Zh. **26** (1985), no. 6, 126–140.
7. Loukas Grafakos and Rodolfo H. Torres, *Multilinear Calderón–Zygmund theory*, Advances in Mathematics **165** (2002), no. 1, 124–164.
8. Tamara G. Kolda and Brett W. Bader, *Tensor decompositions and applications*, SIAM Review **51** (2009), no. 3, 455–500.
9. Joseph B. Kruskal, *Three-way arrays: rank and uniqueness of trilinear decompositions, with application to arithmetic complexity and statistics*, Linear Algebra and its Applications **18** (1977), no. 2, 95–138.
10. Jean-Louis Loday and Bruno Vallette, *Algebraic operads*, Grundlehren der mathematischen Wissenschaften, vol. 346, Springer, 2012.
11. Michael Reed and Barry Simon, *Methods of modern mathematical physics i: Functional analysis*, revised and enlarged ed., Academic Press, 1980.
12. Frank W. Warner, *Foundations of differentiable manifolds and lie groups*, Graduate Texts in Mathematics, vol. 94, Springer, 1983.
13. Donald Yau, *Colored operads*, Graduate Studies in Mathematics, vol. 170, American Mathematical Society, 2016.

Independent Researcher, Apizaco, Tlaxcala, Mexico